Algorithms and Data Structures – Multiple-Choice Test

Instructions: This test consists of 10 multiple-choice questions based on the five lecture materials provided. Each question has four possible answers (A, B, C, D), of which only one is correct.

Student practice quiz

QUESTION 1

Which of the following properties ensures that an algorithm terminates after a finite number of steps?

A) Generality

B) Finiteness

C) Scalability

D) Robustness

QUESTION 2

Which asymptotic notation describes an upper bound on the growth rate of an algorithm's cost function?

A) Big Omega (Ω)

B) Big Theta (Θ)

C) Big O (O)

D) None of the above

QUESTION 3

What is the asymptotic solution of the recurrence $T(n) = 2T(n/2) + \Theta(n)$, according to the Master Theorem?

A) $\Theta(\log n)$

B) $\Theta(n)$

C) $\Theta(n \log n)$

D) $\Theta(n^2)$

QUESTION 4

What is the lower bound on the worst-case number of comparisons required by any comparison-based sorting algorithm?

A) $\Omega(\log n)$

B) $\Omega(n)$

C) $\Omega(n \log n)$

D) $\Omega(n^2)$

QUESTION 5

What is the time complexity of Selection Sort in the best case?

A) $O(1)$

B) $O(n)$

C) $O(n \log n)$

D) $O(n^2)$

QUESTION 6

Which sorting algorithm is particularly effective for small or nearly sorted datasets?

A) Selection Sort

B) Insertion Sort

C) Heap Sort

D) Quick Sort

QUESTION 7

Which statement about Merge Sort is correct?

A) It uses a greedy strategy to select a pivot.

B) Its worst-case time complexity is $O(n^2)$.

C) It recursively divides the array and merges sorted subarrays.

D) It requires the input array to be already sorted.

QUESTION 8

What is the worst-case time complexity of Binary Search on a sorted array of n elements?

A) O(1)

B) O(log n)

C) O(n)

D) O(n log n)

QUESTION 9

Under which condition does Dijkstra's algorithm correctly compute shortest paths?

A) The graph must be acyclic.

B) All edge weights must be non-negative.

C) The graph must be complete.

D) Every edge must have the same weight.

QUESTION 10

Which graph traversal algorithm explores all vertices at the current depth before proceeding to the next depth level?

A) Depth-First Search (DFS)

B) Bellman-Ford

C) Breadth-First Search (BFS)

D) Floyd-Warshall

Question

Correct Answer

1

B – Finiteness

Algorithm properties

2

C – Big O (O)

Asymptotic notation

3

C – Θ(n log n)

Master Theorem

4

C – Ω(n log n)

Comparison-based sorting

5

D – $O(n^2)$

Selection Sort

6

B – Insertion Sort

Sorting algorithms

7

C – Divide and merge sorted subarrays

Merge Sort

8

B – O(log n)

Binary Search

9

B – Non-negative edge weights

Dijkstra's algorithm

10

C – Breadth-First Search (BFS)

Graph traversal